

STAT 516 - Homework 3

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. (12 points) In a company, 12% of employees in their first five years earn \$100,000 or more, and 24% of employees beyond their first five years earn \$100,000 or more. Overall, 15.6% of employees earn \$100,000 or more. Determine the proportion of employees in their first five years.

$$P(100k^+ | \text{first 5 years}) = 0.12$$

$$P(100k^+ | \text{beyond 5 years}) = 0.24$$

$$P(100k^+) = 0.156$$

↓
closer to 0.12 so more people
in the first 5 years

$$0.24 - 0.156 = 0.084$$

$$0.24 - 0.12 = 0.12$$

$$\text{Proportion of first 5 years} = \frac{0.084}{0.12} = 0.70$$

15.6% is 70% of the way from 24% to 12%.

2. (13 points) A disease affects 2% of a population. A medical test is used to detect the disease.

If a person has the disease, the test gives a positive result with probability 90%. If a person does not have the disease, the test gives a positive result with probability 0.1%.

A randomly selected person tests positive. What is the probability that this person actually has the disease?

Let D = Disease, $+$ = Positive Result:

$$P(D) = 0.02 \quad P(D^c) = 1 - 0.02 = 0.98$$

$$P(+|D) = 0.90$$

$$P(+|D^c) = 0.001$$

$$P(D|+) = \frac{P(+|D)P(D)}{P(+|D)P(D) + P(+|D^c)P(D^c)} = \frac{(0.90)(0.02)}{(0.90)(0.02) + (0.001)(0.98)}$$
$$= \frac{0.018}{0.01898}$$

$$P(D|+) = 0.9484$$

3. (10 points) A student takes an examination consisting of true-false questions. If the student knows the answer to a question, then the student answers correctly. Otherwise, the student guesses the answer. If the student does not know the answer, the probability of guessing correctly is 0.5. For a given question, the probability that the student knows the answer, given that the student answered correctly, is 0.75. Find the probability that the student knows the answer to the question. $P(K)$

Let K = Knowing, K^c = Not Knowing (Guessing)
 C = Correct, C^c = Not Correct (Wrong)

$$P(C|K) = 1$$

$$P(C|K^c) = 0.5$$

$$P(K|C) = 0.75$$

$$P(K|C) = \frac{P(C|K)P(K)}{P(C|K)P(K) + P(C|K^c)P(K^c)}$$

$$0.75 = \frac{1 \cdot P(K)}{P(K) + (0.5)(1 - P(K))} = \frac{P(K)}{0.5 + 0.5P(K)}$$

$$0.75(0.5 + 0.5P(K)) = P(K)$$

$$0.375 = 0.625P(K)$$

$$P(K) = 0.60$$

4. (16 points) Find the appropriate value of c for each of the following functions so that each defines a probability mass function on the positive integers 1, 2, 3, $\sum_{x=1}^{\infty} p_X(x) = 1$

(a) $p_X(x) = c 2^{-x}$

(b) $p_X(x) = c \frac{2^x}{x!}$

a) $p_X(x) = c 2^{-x}$

$$\sum_{x=1}^{\infty} c 2^{-x} = 1 \rightarrow c \sum_{x=1}^{\infty} 2^{-x} = 1$$

$$x=1 \rightarrow 1/2$$

$$x=2 \rightarrow 1/4$$

$$x=3 \rightarrow 1/8$$

⋮

• Geometric Series

$$a + ra + r^2a = \frac{a}{1-r}$$

$$a = \frac{1}{2}, r = \frac{1}{2} \rightarrow \frac{1/2}{1-1/2} = 1 = \sum_{x=1}^{\infty} 2^{-x}$$

$$c(1) = 1$$

b) $p_X(x) = c \frac{2^x}{x!}$

• Power Series

$$c \sum_{x=1}^{\infty} \frac{2^x}{x!} = 1$$

$$\sum_{x=0}^{\infty} \frac{x^n}{n!} = e^a$$

$$e^2 = \sum_{x=0}^{\infty} \frac{2^x}{x!}$$

remove $x=0$

$$e^2 - 1 = \sum_{x=1}^{\infty} \frac{2^x}{x!}$$

$$\frac{2^0}{0!} = 1$$

$$c(e^2 - 1) = 1$$

$$c = 1/e^2 - 1$$

5. (10 points) A continuous random variable X has density function

$$f(x) = \begin{cases} \frac{1}{(1+x)^2} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(1 < X < 4)$.

$$P(1 < X < 4) = \int_1^4 \frac{1}{(1+x)^2} dx = \int_1^4 (1+x)^{-2} dx$$

$$= \left[-\frac{1}{1+x} \right]_1^4 = -\frac{1}{5} + \frac{1}{2} = -\frac{2}{10} + \frac{5}{10}$$

plug in bounds

$$P(1 < X < 4) = 3/10$$

6. (12 points) For a random variable X , you are given the PDF =

$$f(x) = \begin{cases} cx^3 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(0.5 \leq X \leq 1.5)$.

$$\int_0^2 cx^3 dx = 1 \rightarrow c \int_0^2 x^3 dx = 1$$

$$c \left[\frac{x^4}{4} \right]_0^2 = 1$$

$$c \left(\frac{2^4}{4} - \frac{0^4}{4} \right) = 1$$

$$c(4) = 1$$

$$c = \frac{1}{4}$$

$$P(0.5 \leq X \leq 1.5) = \int_{0.5}^{1.5} \frac{1}{4} x^3 dx = \frac{1}{4} \left[\frac{x^4}{4} \right]_{0.5}^{1.5}$$

$$= \frac{1}{16} (1.5^4 - 0.5^4)$$

$$= \frac{1}{16} \left(\frac{81}{16} - \frac{1}{16} \right) = \frac{1}{16} \left(\frac{80}{16} \right) = \frac{80}{256}$$

$$P(0.5 \leq X \leq 1.5) = \frac{5}{16}$$

7. (12 points) The distribution of the size of claims paid under an insurance policy has PDF

$$f(x) = \begin{cases} cx^a & 0 < x < 6 \\ 0 & \text{otherwise} \end{cases}$$

where $a > 0$ and $c > 0$. For a randomly selected claim, the probability that the size of the claim is less than 4.5 is 0.50. $P(X < 4.5) = 0.5$

Calculate the probability that the size of the claim is greater than 5. $P(X > 5)$

$$\int_0^6 cx^a dx = 1 \rightarrow c \int_0^6 x^a dx$$

$$c \left[\frac{x^{a+1}}{a+1} \right]_0^6 = 1$$

$$c \left(\frac{6^{a+1}}{a+1} - \frac{0^{a+1}}{a+1} \right) = 1$$

$$c \left(\frac{6^{a+1}}{a+1} \right) = 1 \rightarrow c = \frac{a+1}{6^{a+1}}$$

$$P(X < x) = \int_0^x ct^a dt = c \left[\frac{t^{a+1}}{a+1} \right]_0^x = c \left(\frac{x^{a+1}}{a+1} \right)$$

$$= \frac{c}{a+1} x^{a+1} = \frac{\left(\frac{a+1}{6^{a+1}} \right)}{a+1} x^{a+1} = \frac{x^{a+1}}{6^{a+1}}$$

$$P(X < x) = \left(\frac{x}{6} \right)^{a+1}$$

$$P(X < 4.5) = \left(\frac{4.5}{6} \right)^{a+1} = \left(\frac{3}{4} \right)^{a+1} = 0.5$$

solve for $a+1$: $\log 3/4$ $\log 3/4$

$$a+1 = 2.40942$$

$$P(X < 5) = 1 - P(X > 5) = 1 - \left(\frac{5}{6} \right)^{2.40942}$$

$$P(X < 5) = 0.355... \approx 0.356$$

8. (15 points) The distribution of the amount of time, in hours, for a staff meeting has PDF

$$f(x) = \begin{cases} c(x-1)^5(2-x) & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Calculate the probability that a staff meeting lasts longer than 1.6 hours. $P(X > 1.6)$

To simplify, let $u = x-1$

when $x=1$, $u=0$

$x=2$, $u=1$

$x=1.6$, $u=0.6$

$$c \int_0^1 u^5 (1-u) du = 1$$

$$c \int_0^1 u^5 - u^6 du = 1$$

$$c \left[\frac{u^6}{6} - \frac{u^7}{7} \right]_0^1$$

$$c \left(\frac{1}{6} - \frac{1}{7} \right) = 1$$

$$c \left(\frac{1}{42} \right) = 1$$

$$c = 42$$

$$P(X > 1.6) = 42 \int_{0.6}^1 u^5 (1-u) du = 42 \left[\frac{u^6}{6} - \frac{u^7}{7} \right]_{0.6}^1$$

$$= 42 \left[\left(\frac{1}{6} - \frac{1}{7} \right) - \left(\frac{0.6^6}{6} - \frac{0.6^7}{7} \right) \right]$$

$$P(X > 1.6) = 0.8413696 \approx 0.841$$